

Master on Artificial Intelligence

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Introduction to Human Language Technologies

5. Lexical semantics



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Outline

- 1 Semantics
 - Motivation of lexical semantics
 - Resources
- 2 WordNet
 - Definition
 - Similarities
- 3 SentiWordNet
- 4 Sentiment analysis
 - Definition
 - Examples of methods

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Semantics

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Semantics deals with the meaning:

- Lexical semantics: deals with the meaning of individual words
- Compositional semantics: deals with the construction of meaning usually in high concordance with syntax

This session focuses on lexical semantics

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Semantics
Motivation of
lexical semantics

WordNet

SentiWordNet

Sentiment
analysis

Motivation of lexical semantics

Some examples of usefulness:

- Discovery of semantic patterns

Ex: USA **bombed** Hiroshima

They began to **bombard** the defenses

→ A **sense_12533** B

- Determine discourse relations

Ex: [Anna will show up **later.**] [She has **missed the train.**] →
explanation

Ex: [Mathew is good cooking.] [Albert fails making every dish] →
contrast

- Twitter sentiment analysis

Ex: @vooda1: CNN Declines to Air White House Press Conference
Live YES! THANK YOU @CNN FOR NOT LEGITIMI...
positive

Ex: @Slate: Donald Trump's administration: "Government by the
worst men."
negative

Semantics
Motivation of
lexical semantics

WordNet

SentiWordNet

Sentiment
analysis

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- Resources

2 WordNet

- Definition
- Similarities

3 SentiWordNet

4 Sentiment analysis

- Definition
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Semantics

Resources

WordNet

SentiWordNet

Sentiment
analysis

Resources of lexical semantics

Semantics

Resources

WordNet

SentiWordNet

Sentiment
analysis

- Knowledge-based resources: represented as graphs
 - Ex: **WordNet** (English lexical ontology)
 - SentiWordNet** (sentiment polarity into WordNet)
 - BabelNet (Wikipedia+WordNet)
 - VerbNet (syntactic/semantic verbal behaviour)
 - FrameNet (conceptual behaviour –fine-grained event representation–)
 - ConceptNet (common sense knowledge)
- Corpus-based resources: contextual usage of words
 - Ex: Latent Semantic Analysis (LSA)
 - Word embeddings
 - We will study them in AHLT**

Resources of lexical semantics

Semantics

Resources

WordNet

SentiWordNet

Sentiment
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WordNet	https://wordnet.princeton.edu/
SentiWordNet	http://sentiwordnet.isti.cnr.it/
BabelNet	https://babelnet.org/
VerbNet	https://verbs.colorado.edu/verbnet/
FrameNet	https://framenet.icsi.berkeley.edu/fndrupal/
LSA	accessible from
Word embeddings	https://radimrehurek.com/gensim/

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 - Definition
 - Examples of methods

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Outline

- 1 Semantics
 - Motivation of lexical semantics
 - Resources
- 2 WordNet
 - Definition
 - Similarities
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- 4 Sentiment analysis
 - Definition
 - Examples of methods

Semantics

WordNet

Definition

SentiWordNet

Sentiment
analysis

WordNet

- Free large lexical database of English
- Contains only nouns, verbs, adjectives and adverbs
- Words are grouped into synonyms sets (*synsets*)
- each *synset* has an associated gloss and some examples
- *synsets* are interlinked by means of lexical relations

<http://wordnetweb.princeton.edu/perl/webwn>

Word to search for:

Search WordNet

Display Options:

Change

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations

Display options for sense: (gloss) "an example sentence"

Noun

- **S: (n) age** (how long something has existed) *"it was replaced because of its age"*
- **S: (n) historic period, age** (an era of history having some distinctive feature) *"we live in a litigious age"*

Semantics

WordNet

Definition

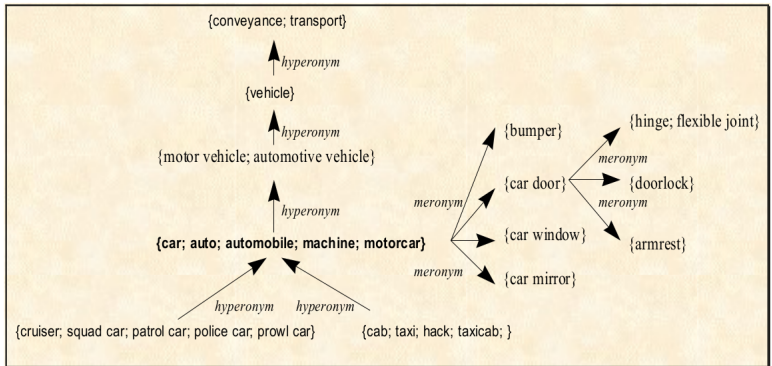
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Sentiment

analysis

Lexical relations

Example of Lexical Relation Net



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WordNet

Definition

SentiWordNet

Sentiment
analysis

Lexical relations

- **Synonym**: same meaning. Ex: age - historic_period
- **Antonym**: opposite meaning. Ex: dark - light
- **Homophone**: same sound. Ex: son - sun
- **Homograph**: same written form. Ex: lead (noun - verb)
- **Polysemy**: different related meaning. Ex: newspaper (paper - firm)
- **Homonymy**: different unrelated meaning. Ex: position (place - status)
- **Hypernym**: parent. Ex: cat - feline
- **Hyponym**: child. Ex: feline - cat
- **Holonymy**: group, whole. Ex: class - student
- **Meronymy**: member, part. Ex: student - class
- **Metonymy**: substitution of entity. Ex: We ordered many delicious dishes at the restaurant.

Semantics

WordNet

Definition

SentiWordNet

Sentiment

analysis

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Semantics

WordNet

Similarities

SentiWordNet

Sentiment
analysis

Similarities in WordNet

- Shortest Path Length: $Sim(s_1, s_2) = \frac{1}{1+SPL(s_1, s_2)}$
where $SPL(s_1, s_2)$ = Shortest Path Length from s_1 to s_2
- Leacock & Chodorow: $Sim(s_1, s_2) = -\log \frac{SPL(s_1, s_2)}{2 \cdot \text{MaxDepth}}$
where $depth(s)$ = depth of s in the ontology
 $MaxDepth = \max_{s \in WN} depth(s)$
- Wu & Palmer: $Sim(s_1, s_2) = \frac{2 \cdot depth(LCS(s_1, s_2))}{depth(s_1) + depth(s_2)}$
where $LCS(s_1, s_2)$ = Lowest Common Subsumer of s_1 and s_2
- Lin: $Sim(s_1, s_2) = \frac{2 \cdot IC(LCS(s_1, s_2))}{IC(s_1) + IC(s_2)}$
where $IC(s) = -\log_2 P(s)$ = information content of s (from frequencies in a corpus)

Semantics

WordNet

Similarities

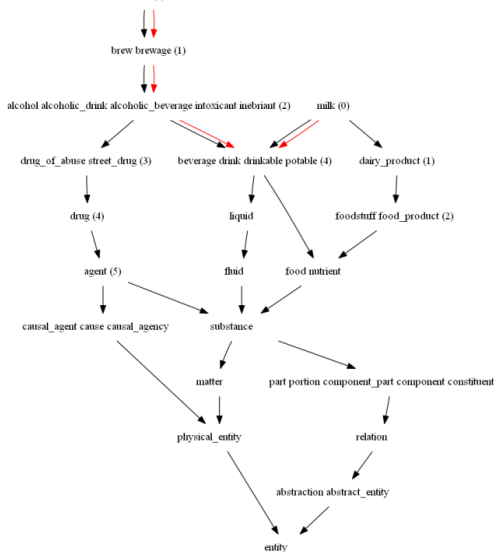
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Sentiment

analysis

Example / exercise

Sandipan Dey (UMBC) beer (0)



$$spl(beer, milk) = 4$$

$$Sim_{spl}(beer, milk) = 0.2$$

$$Sim_{wp}(beer, milk) = 0.71$$

$$Sim_{spl}(drug, milk)?$$

$$Sim_{wp}(drug, milk)?$$

Semantics

WordNet

Similarities

SentiWordNet

Sentiment

analysis

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 - Resources
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Semantics

WordNet

SentiWordNet

Sentiment
analysis

Definition

Extension of wordnet that adds for each synset 3 measures:

- positive_score
- negative_score
- objective_score = 1 - positive_score - negative_score

Wordnet		SentiWordnet		
Antonym Synsets	Gloss	obj	pos	neg
bad.a.01	having undesirable or negative qualities	0.375	0.0	0.625
good.a.01	having desirable or positive qualities. . .	0.25	0.75	0.0
bad.n.01	that which is below standard or expectations as of ethics or decency	0.125	0.0	0.875
good.n.03	that which is pleasing, valuable, useful	0.375	0.625	0.0

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 - Definition
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- 3 SentiWordNet
- 4 Sentiment analysis
 - Definition
 - Examples of methods

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Outline

- 1 Semantics
 - Motivation of lexical semantics
 - Resources
- 2 WordNet
 - Definition
 - Similarities
- 3 SentiWordNet
- 4 Sentiment analysis
 - Definition
 - Examples of methods

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Definition

Sentiment analysis

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Definition

Different subtasks:

- **Opinion detection**: given a piece of text (document or sentence), is it an objective text or a subjective one?
- **Polarity classification**: given a subjective piece of text, is it a positive opinion or a negative one?
- **Opinion extraction**: given a subjective piece of text, recognise the focuses of the opinion (templates <entity, aspect, polarity>).

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 - Definition
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- 3 SentiWordNet
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 - Examples of methods

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Examples of
methods

Unsupervised sentiment analysis

Possible solution:

$$h(D) = \sum_{s \in \hat{D}} \text{score}(s)$$

$\hat{D}$ is usually the set of synsets related to adjectives, or to nouns and adjectives, or to nouns, verbs, adjectives and adverbs.

- Opinion detection:

$$\text{score}(s) = 1 - \text{obj}_s \quad \text{or} \quad \text{score}(s) = \text{obj}_s$$

- Polarity classification:

$$\text{score}(s) = \text{pos}_s - \text{neg}_s$$

Pros:

- no need for training corpora

Cons:

- low results
- need for POS and WSD taggers

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Examples of
methods

Supervised sentiment analysis

Possible solution:

Bag of words with Naïve Bayes

$$h(D) = h(w_1, \dots, w_n) = \underset{y}{\operatorname{argmax}} P(y) \prod_{i=1}^n P(w_i|y)$$

where y is the category (positive/negative, subjective/objective), and $w_1, \dots, w_n$ is the bag of words related to D

- Given a training corpus $C = \{d_i\}$ partitioned into subsets Y_1 and Y_2

- $P(y) \approx P_{MLE}(y) = \frac{|Y_i|}{|C|}$

- $P(w_i|y) \approx P_{MLE}(w_i|Y_j) = \frac{c(w_i, Y_j)}{\sum_{w_i \in Y_j} c(w_i, Y_j)}$

Pros:

- higher results
- no need for POS and WSD taggers

Cons:

- need for training corpora

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Examples of
methods

Hybrid approach for sentiment analysis

Possible solution:

- Combine two supervised methods with SentiWordnet method
- I.e., consensuate the output of the three methods, using *voting*, for instance:
 - if at least 2 of the methods answer y then output y
 - else output the answer of the method with better accuracy in the training corpus

The combination improves the results of the isolated methods

Semantics

WordNet

SentiWordNet

Sentiment
analysis

Examples of
methods

Annex

- Base on the Bayes' theorem:

$$P(y|x_1, \dots, x_n) = \frac{P(y)P(x_1, \dots, x_n|y)}{P(x_1, \dots, x_n)}$$

- Naïve assumption of independence between features:

$$P(y|x_1, \dots, x_n) \approx P(y) \prod_{i=1}^n P(x_i|y)$$

- *Maximum likelihood estimation* of $P(y)$ and $P(x_i|y)$ as training model
- Test prediction as:

$$h(x_1, \dots, x_n) = \underset{y}{\operatorname{argmax}} P(y) \prod_{i=1}^n P(x_i|y)$$

- Need a smoothing technique to avoid zero counts:
in NLTK never seen features are discarded